

SCOUTING FOOTBALLERS

AN UNSUPERVISED LEARNING APPROACH

CSML-1000 | By GROUP 3 (*Jean-Jacques Rousseau, David Geller, Kwangjin Park, Petr Kocourek and Puneeth Nagarajaiah*)



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1. Introduction

How much would it cost a soccer team to sign a full roster of players? In English Premier League (EPL), Manchester United FC is the first overall in total salary expense of players, totalling at £227,081,000. On the other hand, Brentford FC ranked last in total salary expense at £15,776,000. In 2021~2022 EPL season, Manchester United ended the season in sixth place, while Brentford ended the season in 13th place. Although Manchester United spent nearly 1339% more on player salary than Brentford FC, it is hard to justify having to

spend that much money on their disappointing season. If paying more for the players doesn't guarantee a good team, then what could be the factors?

To stay competitive in the league, soccer teams must examine the players they need by scouring all over leagues. Agents from each team will look at each player's past performances, techniques, age, physical attributes, game intelligence, mindset, etc to determine if he is the player they are looking for. Scouting for players manually through analysis of individual agent has become more challenging as the number of professional players increased.

If agents can use the help of machine learning that has been trained on a dataset of all available players, they simply need to supply attributes that they find most important to narrow down players of interest. This will greatly save scouts time and money.

- **Business Problem:** Using Machine Learning (Clustering), find the player that meets the specific requirements supplied by a soccer agent. For example, if an agent comes to our web app and enters scores they want for each skill using the slider (eg. choosing a value between 0 and 100), our webapp will output a list of names of players with highest match scores. This will greatly narrow down the number of players the agent has to scout for.
- **Dataset:** FIFA 22 complete player dataset (<https://www.kaggle.com/datasets/stefanoleone992/fifa-22-complete-player-dataset/discussion/360882>)

2. Installation

In this section, we install pycaret and the Kaggle dataset. This section is similar to Assignment 1.

- **Setting Up Modules:** How to install pycaret, kaggle and other required modules
- **Getting Data:** How to import data from Kaggle website directly

```
! pip install pycaret
```

```
! pip install pandas
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
%matplotlib inline
import seaborn as sns; sns.set_theme()
```

```
! pip install Kaggle
! mkdir ~/.kaggle
! cp kaggle.json ~/.kaggle/
! chmod 600 ~/.kaggle/kaggle.json
! kaggle datasets download stefanoleone992/fifa-22-complete-player-dataset
! unzip fifa-22-complete-player-dataset.zip
df = pd.read_csv('/content/players_22.csv')
```

3. General Dataset Analysis

This section explores the characteristics of the data so that we can develop and intuitive feel for the dataset.

3.1 Data Exploration

Data dimension

```
df.shape
```

```
(19239, 110)
```

The dataset has over 19000 players with 110 attributes.

Descriptive Statistics

```
df.describe()
```

	sofifa_id	overall	potential	value_eur
wage_eur \				
count	19239.000000	19239.000000	19239.000000	1.916500e+04
19178.000000				
mean	231468.086959	65.772182	71.079370	2.850452e+06
9017.989363				
std	27039.717497	6.880232	6.086213	7.613700e+06
19470.176724				
min	41.000000	47.000000	49.000000	9.000000e+03
500.000000				
25%	214413.500000	61.000000	67.000000	4.750000e+05
1000.000000				
50%	236543.000000	66.000000	71.000000	9.750000e+05
3000.000000				
75%	253532.500000	70.000000	75.000000	2.000000e+06
8000.000000				
max	264640.000000	93.000000	95.000000	1.940000e+08
350000.000000				

	age	height_cm	weight_kg	club_team_id
league_level \				
count	19239.000000	19239.000000	19239.000000	19178.000000
19178.000000				
mean	25.210822	181.299704	74.943032	50580.498123
1.354364				
std	4.748235	6.863179	7.069434	54401.868535
0.747865				
min	16.000000	155.000000	49.000000	1.000000
1.000000				
25%	21.000000	176.000000	70.000000	479.000000
1.000000				
50%	25.000000	181.000000	75.000000	1938.000000
1.000000				
75%	29.000000	186.000000	80.000000	111139.000000

1.000000				
max	54.000000	206.000000	110.000000	115820.000000
5.000000				

	...	mentality_composure	defending_marking_awareness	\
count	...	19239.000000	19239.000000	
mean	...	57.929830	46.601746	
std	...	12.159326	20.200807	
min	...	12.000000	4.000000	
25%	...	50.000000	29.000000	
50%	...	59.000000	52.000000	
75%	...	66.000000	63.000000	
max	...	96.000000	93.000000	

		defending_standing_tackle	defending_sliding_tackle	\
count		19239.000000	19239.000000	
mean		48.045584	45.906700	
std		21.232718	20.755683	
min		5.000000	5.000000	
25%		28.000000	25.000000	
50%		56.000000	53.000000	
75%		65.000000	63.000000	
max		93.000000	92.000000	

	goalkeeping_diving	goalkeeping_handling	
goalkeeping_kicking	\		
count	19239.000000	19239.000000	19239.000000
mean	16.406102	16.192474	16.055356
std	17.574028	16.839528	16.564554
min	2.000000	2.000000	2.000000
25%	8.000000	8.000000	8.000000
50%	11.000000	11.000000	11.000000
75%	14.000000	14.000000	14.000000
max	91.000000	92.000000	93.000000

	goalkeeping_positioning	goalkeeping_reflexes
goalkeeping_speed		
count	19239.000000	19239.000000
2132.000000		
mean	16.229274	16.491814
36.439962		

std	17.059779	17.884833
10.751563		
min	2.000000	2.000000
15.000000		
25%	8.000000	8.000000
27.000000		
50%	11.000000	11.000000
36.000000		
75%	14.000000	14.000000
45.000000		
max	92.000000	90.000000
65.000000		

[8 rows x 60 columns]

Here you can see general statistic of the data, such as count, mean, min and max. For example, the mean age of the players is 25.2 years old, with the youngest play being 16 years old.

```
pd.set_option('display.max_columns', None)
```

```
type(df)
```

```
pandas.core.frame.DataFrame
```

```
df.head()
```

	sofifa_id	player_url	
0	158023	https://sofifa.com/player/158023/lionel-messi/...	
1	188545	https://sofifa.com/player/188545/robert-lewand...	
2	20801	https://sofifa.com/player/20801/c-ronaldo-dos-...	
3	190871	https://sofifa.com/player/190871/neymar-da-sil...	
4	192985	https://sofifa.com/player/192985/kevin-de-bruy...	

	short_name	long_name	
0	L. Messi	Lionel Andrés Messi Cuccittini	RW,
1	R. Lewandowski	Robert Lewandowski	
2	Cristiano Ronaldo	Cristiano Ronaldo dos Santos Aveiro	
3	Neymar Jr	Neymar da Silva Santos Júnior	LW,
4	K. De Bruyne	Kevin De Bruyne	CM,

	overall	potential	value_eur	wage_eur	age	dob
0	93	93	78000000.0	320000.0	34	1987-06-24
1	92	92	119500000.0	270000.0	32	1988-08-21

2	91	91	45000000.0	270000.0	36	1985-02-05
187						
3	91	91	129000000.0	270000.0	29	1992-02-05
175						
4	91	91	125500000.0	350000.0	30	1991-06-28
181						

	weight_kg	club_team_id	club_name	league_name \
0	72	73.0	Paris Saint-Germain	French Ligue
1				
1	81	21.0	FC Bayern München	German 1.
				Bundesliga
2	83	11.0	Manchester United	English Premier
				League
3	68	73.0	Paris Saint-Germain	French Ligue
1				
4	70	10.0	Manchester City	English Premier
				League

	league_level	club_position	club_jersey_number	club_loaned_from \
0	1.0	RW	30.0	NaN
1	1.0	ST	9.0	NaN
2	1.0	ST	7.0	NaN
3	1.0	LW	10.0	NaN
4	1.0	RCM	17.0	NaN

	club_joined	club_contract_valid_until	nationality_id
nationality_name \			
0	2021-08-10	2023.0	52
			Argentina
1	2014-07-01	2023.0	37
			Poland
2	2021-08-27	2023.0	38
			Portugal
3	2017-08-03	2025.0	54
			Brazil
4	2015-08-30	2025.0	7
			Belgium

	nation_team_id	nation_position	nation_jersey_number	preferred_foot
\				
0	1369.0	RW	10.0	Left
1	1353.0	RS	9.0	Right
2	1354.0	ST	7.0	Right
3	NaN	NaN	NaN	Right

4	1325.0	RCM	7.0	Right
---	--------	-----	-----	-------

	weak_foot	skill_moves	international_reputation	work_rate
body_type \				
0	4	4	5	Medium/Low
Unique				
1	4	4	5	High/Medium
Unique				
2	4	5	5	High/Low
Unique				
3	5	5	5	High/Medium
Unique				
4	5	4	4	High/High
Unique				

	real_face	release_clause_eur \
0	Yes	144300000.0
1	Yes	197200000.0
2	Yes	83300000.0
3	Yes	238700000.0
4	Yes	232200000.0

	player_tags \
0	#Dribbler, #Distance Shooter, #FK Specialist, ...
1	#Aerial Threat, #Distance Shooter, #Clinical F...
2	#Aerial Threat, #Dribbler, #Distance Shooter, ...
3	#Speedster, #Dribbler, #Playmaker, #FK Special...
4	#Dribbler, #Playmaker, #Engine, #Distance Shoo...

	player_traits	pace	shooting
passing \			
0	Finesse Shot, Long Shot Taker (AI), Playmaker ...	85.0	92.0
91.0			
1	Solid Player, Finesse Shot, Outside Foot Shot,...	78.0	92.0
79.0			
2	Power Free-Kick, Flair, Long Shot Taker (AI), ...	87.0	94.0
80.0			
3	Injury Prone, Flair, Speed Dribbler (AI), Play...	91.0	83.0
86.0			
4	Injury Prone, Leadership, Early Crosser, Long ...	76.0	86.0
93.0			

	dribbling	defending	physic	attacking_crossing
attacking_finishing \				
0	95.0	34.0	65.0	85
95				
1	86.0	44.0	82.0	71

95				
2	88.0	34.0	75.0	87
95				
3	94.0	37.0	63.0	85
83				
4	88.0	64.0	78.0	94
82				

	attacking_heading_accuracy	attacking_short_passing
attacking_volleys \		
0	70	91
88		
1	90	85
89		
2	90	80
86		
3	63	86
86		
4	55	94
82		

	skill_dribbling	skill_curve	skill_fk_accuracy	skill_long_passing
\				
0	96	93	94	91
1	85	79	85	70
2	88	81	84	77
3	95	88	87	81
4	88	85	83	93

	skill_ball_control	movement_acceleration	movement_sprint_speed	\
0	96	91	80	
1	88	77	79	
2	88	85	88	
3	95	93	89	
4	91	76	76	

	movement_agility	movement_reactions	movement_balance
power_shot_power \			
0	91	94	95
86			
1	77	93	82
90			
2	86	94	74
94			

3	96	89	84
80			
4	79	91	78
91			

	power_jumping	power_stamina	power_strength	power_long_shots	\
0	68	72	69	94	
1	85	76	86	87	
2	95	77	77	93	
3	64	81	53	81	
4	63	89	74	91	

	mentality_aggression	mentality_interceptions
mentality_positioning	\	
0	44	40
93		
1	81	49
95		
2	63	29
95		
3	63	37
86		
4	76	66
88		

	mentality_vision	mentality_penalties	mentality_composure	\
0	95	75	96	
1	81	90	88	
2	76	88	95	
3	90	93	93	
4	94	83	89	

	defending_marking_awareness	defending_sliding_tackle	\
0	20	35	
1	35	42	
2	24	32	
3	35	32	
4	68	65	

	defending_sliding_tackle	goalkeeping_diving	goalkeeping_handling	\
0	24	6	11	
1	19	15	6	
2	24	7	11	
3	29	9	9	

4	53	15	13
---	----	----	----

	goalkeeping_kicking	goalkeeping_positioning	goalkeeping_reflexes
\			
0	15	14	8
1	12	8	10
2	15	14	11
3	15	15	11
4	5	10	13

	goalkeeping_speed	ls	st	rs	lw	lf	cf	rf	rw	lam	cam
ram \											
0	NaN	89+3	89+3	89+3	92	93	93	93	92	93	93
93											
1	NaN	90+2	90+2	90+2	85	88	88	88	85	86+3	86+3
86+3											
2	NaN	90+1	90+1	90+1	88	89	89	89	88	86+3	86+3
86+3											
3	NaN	83+3	83+3	83+3	90	88	88	88	90	89+2	89+2
89+2											
4	NaN	83+3	83+3	83+3	88	87	87	87	88	89+2	89+2
89+2											

	lm	lcm	cm	rcm	rm	lwb	ldm	cdm	rdm	rwb	lb
lcb \											
0	91+2	87+3	87+3	87+3	91+2	66+3	64+3	64+3	64+3	66+3	61+3
50+3											
1	84+3	80+3	80+3	80+3	84+3	64+3	66+3	66+3	66+3	64+3	61+3
60+3											
2	86+3	78+3	78+3	78+3	86+3	63+3	59+3	59+3	59+3	63+3	60+3
53+3											
3	89+2	82+3	82+3	82+3	89+2	67+3	63+3	63+3	63+3	67+3	62+3
50+3											
4	89+2	89+2	89+2	89+2	89+2	79+3	80+3	80+3	80+3	79+3	75+3
69+3											

	cb	rcb	rb	gk
player_face_url \				
0	50+3	50+3	61+3	19+3
	https://cdn.sofifa.net/players/158/023/22_120.png			
1	60+3	60+3	61+3	19+3
	https://cdn.sofifa.net/players/188/545/22_120.png			
2	53+3	53+3	60+3	20+3

```

https://cdn.sofifa.net/players/020/801/22_120.png
3  50+3  50+3  62+3  20+3
https://cdn.sofifa.net/players/190/871/22_120.png
4  69+3  69+3  75+3  21+3
https://cdn.sofifa.net/players/192/985/22_120.png

```

```

                                club_logo_url \
0  https://cdn.sofifa.net/teams/73/60.png
1  https://cdn.sofifa.net/teams/21/60.png
2  https://cdn.sofifa.net/teams/11/60.png
3  https://cdn.sofifa.net/teams/73/60.png
4  https://cdn.sofifa.net/teams/10/60.png

```

```

                                club_flag_url \
0      https://cdn.sofifa.net/flags/fr.png
1      https://cdn.sofifa.net/flags/de.png
2  https://cdn.sofifa.net/flags/gb-eng.png
3      https://cdn.sofifa.net/flags/fr.png
4  https://cdn.sofifa.net/flags/gb-eng.png

```

```

                                nation_logo_url \
0  https://cdn.sofifa.net/teams/1369/60.png
1  https://cdn.sofifa.net/teams/1353/60.png
2  https://cdn.sofifa.net/teams/1354/60.png
3                                     NaN
4  https://cdn.sofifa.net/teams/1325/60.png

```

```

                                nation_flag_url
0  https://cdn.sofifa.net/flags/ar.png
1  https://cdn.sofifa.net/flags/pl.png
2  https://cdn.sofifa.net/flags/pt.png
3  https://cdn.sofifa.net/flags/br.png
4  https://cdn.sofifa.net/flags/be.png

```

Soccer player requires more than just some technical skills to be an excellent player. For example, Lionel Messi is an excellent ball handler and scorer, but he is also a leader who fosters communication and teamwork within the team, which helps other players perform better as well.

We have expanded column-wise to see all columns. Here, just by taking a quick glance, most of columns are numerical in nature, where different skills are given a score between 0 and 100.

For example, Cristiano Ronaldo has the overall score of 91, shooting score of 94, dribbling score of 88, but defending score of 34.

Checking for missing values

```
df.isnull().sum()
```

```

sofifa_id           0
player_url          0

```

```

short_name          0
long_name           0
player_positions    0
...
player_face_url     0
club_logo_url       61
club_flag_url       61
nation_logo_url     18480
nation_flag_url     0
Length: 110, dtype: int64

```

We also checked if there are any null values, and we saw that much of values for nation_flag_url are missing.

Count numbers

df.select_dtypes(include = ['number']).head() # 50 of the 110 columns contain strings

```

    sofifa_id  overall  potential  value_eur  wage_eur  age
height_cm \
0    158023      93      93    78000000.0   320000.0   34
170
1    188545      92      92   119500000.0   270000.0   32
185
2    20801      91      91    45000000.0   270000.0   36
187
3    190871      91      91   129000000.0   270000.0   29
175
4    192985      91      91   125500000.0   350000.0   30
181

```

```

    weight_kg  club_team_id  league_level  club_jersey_number \
0          72          73.0           1.0           30.0
1          81          21.0           1.0           9.0
2          83          11.0           1.0           7.0
3          68          73.0           1.0          10.0
4          70          10.0           1.0          17.0

```

```

    club_contract_valid_until  nationality_id  nation_team_id \
0                2023.0                52        1369.0
1                2023.0                37        1353.0
2                2023.0                38        1354.0
3                2025.0                54           NaN
4                2025.0                 7        1325.0

```

```

    nation_jersey_number  weak_foot  skill_moves
international_reputation \
0                10.0           4           4
5
1                9.0           4           4
5

```

2	7.0	4	5
5			
3	NaN	5	5
5			
4	7.0	5	4
4			

	release_clause_eur	pace	shooting	passing	dribbling	defending
physic \						
0	144300000.0	85.0	92.0	91.0	95.0	34.0
65.0						
1	197200000.0	78.0	92.0	79.0	86.0	44.0
82.0						
2	83300000.0	87.0	94.0	80.0	88.0	34.0
75.0						
3	238700000.0	91.0	83.0	86.0	94.0	37.0
63.0						
4	232200000.0	76.0	86.0	93.0	88.0	64.0
78.0						

	attacking_crossing	attacking_finishing	attacking_heading_accuracy
\			
0	85	95	70
1	71	95	90
2	87	95	90
3	85	83	63
4	94	82	55

	attacking_short_passing	attacking_volleys	skill_dribbling
skill_curve \			
0	91	88	96
93			
1	85	89	85
79			
2	80	86	88
81			
3	86	86	95
88			
4	94	82	88
85			

	skill_fk_accuracy	skill_long_passing	skill_ball_control
\			
0	94	91	96
1	85	70	88

2	84	77	88
3	87	81	95
4	83	93	91

	movement_acceleration	movement_sprint_speed	movement_agility \
0	91	80	91
1	77	79	77
2	85	88	86
3	93	89	96
4	76	76	79

	movement_reactions	movement_balance	power_shot_power
power_jumping \			
0	94	95	86
68			
1	93	82	90
85			
2	94	74	94
95			
3	89	84	80
64			
4	91	78	91
63			

	power_stamina	power_strength	power_long_shots
mentality_aggression \			
0	72	69	94
44			
1	76	86	87
81			
2	77	77	93
63			
3	81	53	81
63			
4	89	74	91
76			

	mentality_interceptions	mentality_positioning	mentality_vision \
0	40	93	95
1	49	95	81
2	29	95	76
3	37	86	90
4	66	88	94

	mentality_penalties	mentality_composure
defending_marking_awareness \		
0	75	96
20		
1	90	88
35		

2	88	95
24		
3	93	93
35		
4	83	89
68		

	defending_standing_tackle	defending_sliding_tackle
goalkeeping_diving \		
0	35	24
6		
1	42	19
15		
2	32	24
7		
3	32	29
9		
4	65	53
15		

	goalkeeping_handling	goalkeeping_kicking	goalkeeping_positioning
\			
0	11	15	14
1	6	12	8
2	11	15	14
3	9	15	15
4	13	5	10

	goalkeeping_reflexes	goalkeeping_speed
0	8	NaN
1	10	NaN
2	11	NaN
3	11	NaN
4	13	NaN

```
# Count objects (strings)
df.select_dtypes(include = ['object']).head() # 50 of the 110 columns
contain strings
```

	short_name	player_url
0	https://sofifa.com/player/158023/lionel-messi/...	L. Messi
1	https://sofifa.com/player/188545/robert-lewand...	R. Lewandowski

2 <https://sofifa.com/player/20801/c-ronaldo-dos-...> Cristiano Ronaldo
3 <https://sofifa.com/player/190871/neymar-da-sil...> Neymar Jr
4 <https://sofifa.com/player/192985/kevin-de-bruy...> K. De Bruyne

	long_name	player_positions	dob	\
0	Lionel Andrés Messi Cuccittini	RW, ST, CF	1987-06-24	
1	Robert Lewandowski	ST	1988-08-21	
2	Cristiano Ronaldo dos Santos Aveiro	ST, LW	1985-02-05	
3	Neymar da Silva Santos Júnior	LW, CAM	1992-02-05	
4	Kevin De Bruyne	CM, CAM	1991-06-28	

	club_name	league_name	club_position
0	Paris Saint-Germain	French Ligue 1	RW
1	FC Bayern München	German 1. Bundesliga	ST
2	Manchester United	English Premier League	ST
3	Paris Saint-Germain	French Ligue 1	LW
4	Manchester City	English Premier League	RCM

	club_joined	nationality_name	nation_position	preferred_foot
0	2021-08-10	Argentina	RW	Left
1	2014-07-01	Poland	RS	Right
2	2021-08-27	Portugal	ST	Right
3	2017-08-03	Brazil	NaN	Right
4	2015-08-30	Belgium	RCM	Right

	body_type	real_face
0	Unique Specialist, ...	Yes #Dribbler, #Distance Shooter, #FK
1	Unique F...	Yes #Aerial Threat, #Distance Shooter, #Clinical
2	Unique Shooter, ...	Yes #Aerial Threat, #Dribbler, #Distance
3	Unique Special...	Yes #Speedster, #Dribbler, #Playmaker, #FK

4 Unique Yes #Dribbler, #Playmaker, #Engine, #Distance
Shoo...

	player_traits	ls	st	rs
lw \				
0	Finesse Shot, Long Shot Taker (AI), Playmaker ...	89+3	89+3	89+3
92				
1	Solid Player, Finesse Shot, Outside Foot Shot,...	90+2	90+2	90+2
85				
2	Power Free-Kick, Flair, Long Shot Taker (AI), ...	90+1	90+1	90+1
88				
3	Injury Prone, Flair, Speed Dribbler (AI), Play...	83+3	83+3	83+3
90				
4	Injury Prone, Leadership, Early Crosser, Long ...	83+3	83+3	83+3
88				

	lf	cf	rf	rw	lam	cam	ram	lm	lcm	cm	rcm	rm
lwb \												
0	93	93	93	92	93	93	93	91+2	87+3	87+3	87+3	91+2
66+3		64+3										
1	88	88	88	85	86+3	86+3	86+3	84+3	80+3	80+3	80+3	84+3
64+3		66+3										
2	89	89	89	88	86+3	86+3	86+3	86+3	78+3	78+3	78+3	86+3
63+3		59+3										
3	88	88	88	90	89+2	89+2	89+2	89+2	82+3	82+3	82+3	89+2
67+3		63+3										
4	87	87	87	88	89+2	89+2	89+2	89+2	89+2	89+2	89+2	89+2
79+3		80+3										

	cdm	rdm	rwb	lb	lcb	cb	rcb	rb	gk	\
0	64+3	64+3	66+3	61+3	50+3	50+3	50+3	61+3	19+3	
1	66+3	66+3	64+3	61+3	60+3	60+3	60+3	61+3	19+3	
2	59+3	59+3	63+3	60+3	53+3	53+3	53+3	60+3	20+3	
3	63+3	63+3	67+3	62+3	50+3	50+3	50+3	62+3	20+3	
4	80+3	80+3	79+3	75+3	69+3	69+3	69+3	75+3	21+3	

	player_face_url	\
0	https://cdn.sofifa.net/players/158/023/22_120.png	
1	https://cdn.sofifa.net/players/188/545/22_120.png	
2	https://cdn.sofifa.net/players/020/801/22_120.png	
3	https://cdn.sofifa.net/players/190/871/22_120.png	
4	https://cdn.sofifa.net/players/192/985/22_120.png	

	club_logo_url	\
0	https://cdn.sofifa.net/teams/73/60.png	
1	https://cdn.sofifa.net/teams/21/60.png	
2	https://cdn.sofifa.net/teams/11/60.png	
3	https://cdn.sofifa.net/teams/73/60.png	
4	https://cdn.sofifa.net/teams/10/60.png	

```

club_flag_url \
0 https://cdn.sofifa.net/flags/fr.png
1 https://cdn.sofifa.net/flags/de.png
2 https://cdn.sofifa.net/flags/gb-eng.png
3 https://cdn.sofifa.net/flags/fr.png
4 https://cdn.sofifa.net/flags/gb-eng.png

nation_logo_url \
0 https://cdn.sofifa.net/teams/1369/60.png
1 https://cdn.sofifa.net/teams/1353/60.png
2 https://cdn.sofifa.net/teams/1354/60.png
3 NaN
4 https://cdn.sofifa.net/teams/1325/60.png

nation_flag_url
0 https://cdn.sofifa.net/flags/ar.png
1 https://cdn.sofifa.net/flags/pl.png
2 https://cdn.sofifa.net/flags/pt.png
3 https://cdn.sofifa.net/flags/br.png
4 https://cdn.sofifa.net/flags/be.png

```

We wanted to see how many of our columns have string datatype and how many of our columns are numerical datatype.

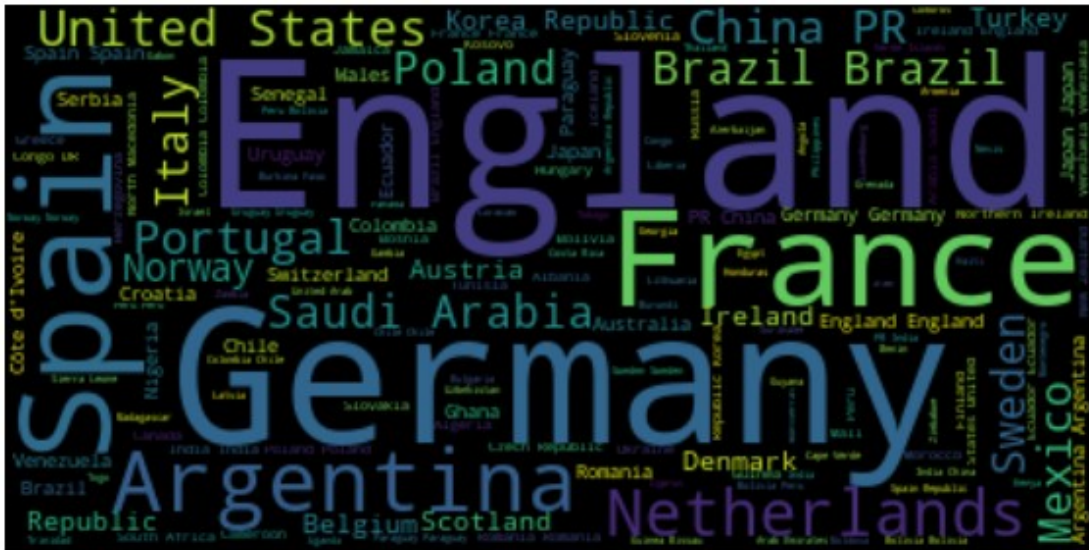
Here, we can see that 60 out of 110 features are numerical, which means 50 out of 110 features are strings.

Most popular nationalities in the dataset

```

from wordcloud import WordCloud
nationality_name = " ".join(n for n in df['nationality_name'])
plt.figure(figsize=(10, 10))
wc = WordCloud().generate(nationality_name)
plt.imshow(wc, interpolation='bilinear')
plt.axis('off')
plt.show()

```



Using WorldCloud, here we see that most of players in the dataset are from England, Germany, France and Spain.

General information about this dataset:

1. The dataset include the players data for the Career Mode from FIFA 22
2. There are over 100 attributes
3. It has a mix of text and numerical values (60 numerical attributes and 50 text attributes)
4. The dataset shows player positions, with the role in the club and in the national team
5. The dataset shows player attributes with statistics as Attacking, Skills, Defense, Mentality, GK Skills, etc
6. The dataset shows player personal data like Nationality, Club, DateOfBirth, Wage, Salary, etc.
7. URL of the uploaded player faces, club and nation logos
8. Most of null data is from nation_flag_url
9. Most players in the dataset are from England, Germany, France and Spain

3.2 In-depth Data Exploration - Conditional Selection

Potential describes how much a certain player is expected to grow in their career.

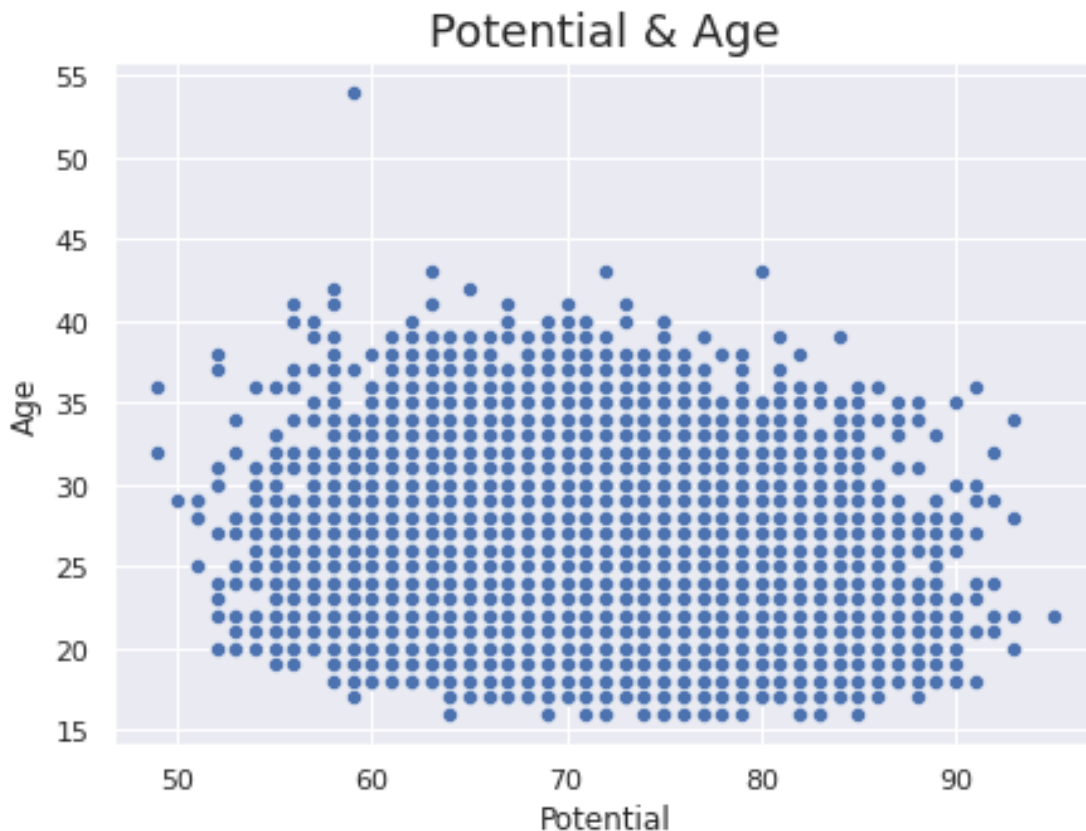
```
# Player potential versus wages
plt.figure(figsize=(7, 5))
ax = sns.scatterplot(x =df['potential'], y = df['wage_eur'])
plt.xlabel("Potential")
plt.ylabel("Wage EUR")
plt.title("Potential & wage", fontsize = 18)
plt.show()
```



Here we see that, players with higher potential generally earns more wages. For example, Lionel Messi and Robert Lewandowski both has two of highest potential in this dataset, and both earn highest wages as well.

Player potential versus age

```
plt.figure(figsize=(7, 5))
ax = sns.scatterplot(x =df['potential'], y = df['age'])
plt.xlabel("Potential")
plt.ylabel("Age")
plt.title("Potential & Age", fontsize = 18)
plt.show()
```



Here, we don't see a clear correlation between potential and age. This possibly means that players age is not a factor in determining that player's potential. It is possible that younger players are generally healthier and has better physical attributes than older players. But older players have experience, game intelligence and techniques to overcome physical disadvantage to younger players.

The game of soccer is a complex game that requires both physical fitness and skills, so regardless of age, most players continue to play at high level.

Here are the 10 oldest players. The outlier above is K. Miura from Japan at 54 years old. He is the oldest player in the dataset.

```
oldest_players = df[['short_name',
                    'nationality_name',
                    'age',
                    'potential',
                    ]].nlargest(10, ['age']).set_index('short_name')
```

oldest_players

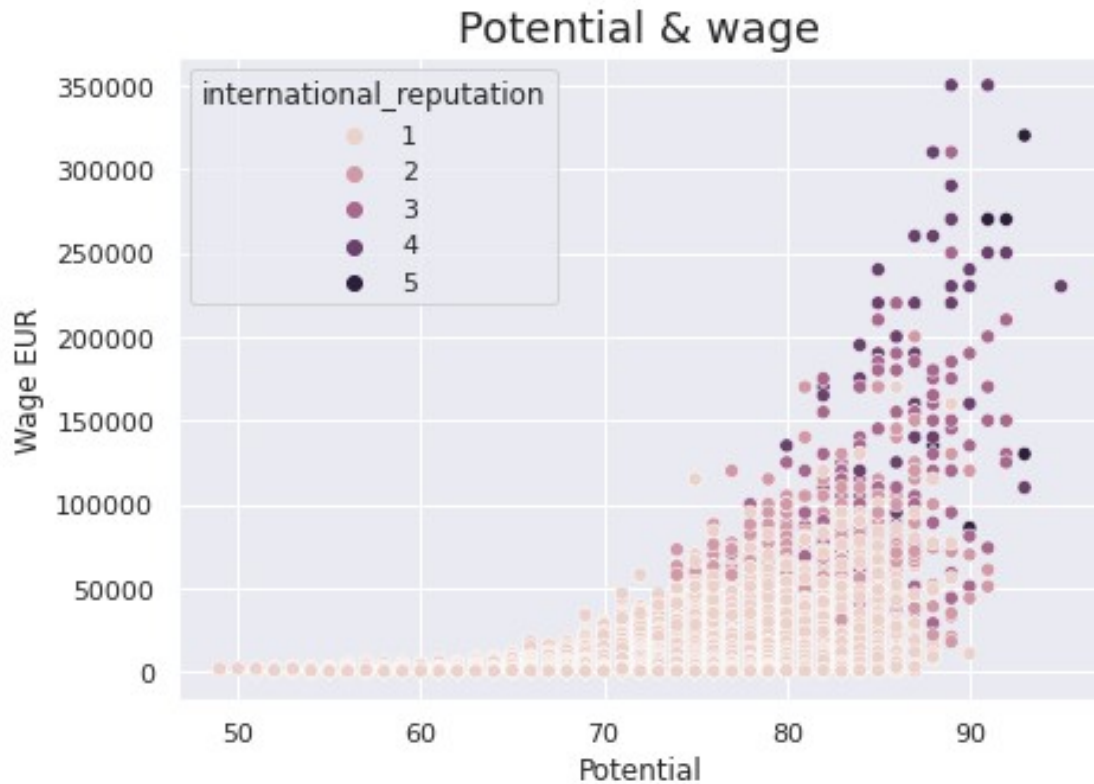
short_name	nationality_name	age	potential
K. Miura	Japan	54	59
G. Buffon	Italy	43	80
C. Lucchetti	Argentina	43	72
S. Nakamura	Japan	43	63
D. Vaca	Bolivia	42	65
K. Ellison	England	42	58
S. Torrico	Argentina	41	73
A. Boruc	Poland	41	70
P. Da Silva	Paraguay	41	67
S. Lukić	Bosnia and Herzegovina	41	63



Here we see players who are oldest in the dataset. We see that K. Miura is the oldest player at age 54, while G. Buffon is the second oldest player at age 43. We see that Gianluigi Buffon continues to play at high level even at older age compared to other older players.

Player potential versus wages, filtered by International reputation

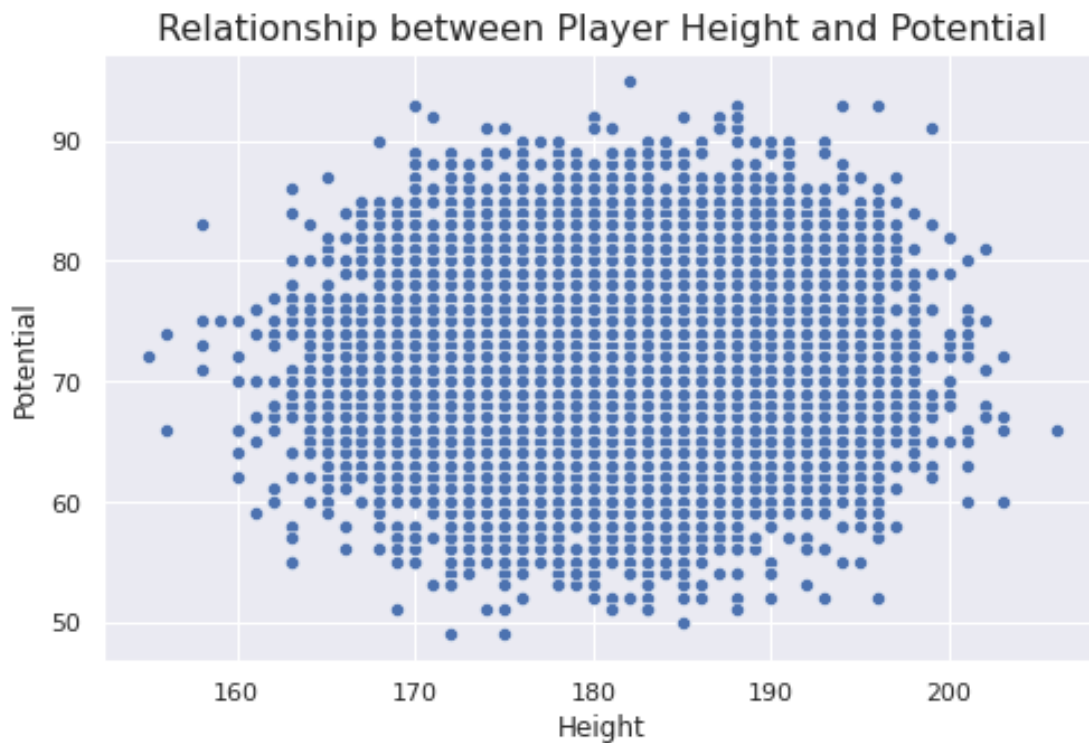
```
plt.figure(figsize=(7, 5))
ax = sns.scatterplot(x=df['potential'], y=df['wage_eur'], hue=df['international_reputation'])
plt.xlabel("Potential")
plt.ylabel("Wage EUR")
plt.title("Potential & wage", fontsize=18)
plt.show()
```



This chart is similar to the one that shows correlation between Potential and Wage, but we added a third factor called International Reputation. We see that players with highest potential earns high wages, but also has high international reputation.

For example, players like Cristiano Ronaldo and Lionel Messi would fit this category of players with high_potential/high_wage/high_reputation.

```
# Player height versus potential
fig, ax = plt.subplots(figsize = (8,5))
ax = sns.scatterplot(x =df['height_cm'], y = df['potential'])
plt.xlabel("Height")
plt.ylabel("Potential")
plt.title("Relationship between Player Height and Potential", fontsize
= 16)
plt.show()
```



Here, we see the chart showing the correlation between player's height and their potential. We don't see a clear correlation between two attributes, as the game of soccer can be played by players of all heights.

3.3 Pandas Profiling Feature Analysis

```
# List all features in dataset
```

```
df.columns
```

```
Index(['sofifa_id', 'player_url', 'short_name', 'long_name',
      'player_positions', 'overall', 'potential', 'value_eur',
      'wage_eur',
      'age',
      ...,
      'lcb', 'cb', 'rcb', 'rb', 'gk', 'player_face_url',
      'club_logo_url',
      'club_flag_url', 'nation_logo_url', 'nation_flag_url'],
      dtype='object', length=110)
```

There are over 100 attributes in this dataset. Obviously, training an algorithm would be cumbersome, takes long time and wouldn't produce clear interpretation of our dataset.

This is why we need to narrow down to a small number of features that we are interested in to build an accurate Machine Learning Model. We will be using Pandas Profiling to get a better picture of the features.

In this section, our data analysis is on the entire dataset, but in '**Model Analysis**' section, we have narrowed down to Goalkeeper position and a small number of attributes.

Pick out the features of interest

```
df_subset = df[['short_name', 'player_positions', 'overall',  
'potential', 'wage_eur', 'age', 'club_name', 'nationality_name',  
'skill_moves', 'work_rate', 'shooting', 'passing', 'defending']]
```

```
type(df_subset)
```

```
pandas.core.frame.DataFrame
```

```
df_subset.describe()
```

	overall	potential	wage_eur	age
count	19239.000000	19239.000000	19178.000000	19239.000000
mean	65.772182	71.079370	9017.989363	25.210822
std	6.880232	6.086213	19470.176724	4.748235
min	47.000000	49.000000	500.000000	16.000000
25%	61.000000	67.000000	1000.000000	21.000000
50%	66.000000	71.000000	3000.000000	25.000000
75%	70.000000	75.000000	8000.000000	29.000000
max	93.000000	95.000000	350000.000000	54.000000

	shooting	passing	defending
count	17107.000000	17107.000000	17107.000000
mean	52.345297	57.312562	51.703630
std	14.051623	10.068965	16.189746
min	18.000000	25.000000	14.000000
25%	42.000000	51.000000	37.000000
50%	54.000000	58.000000	56.000000
75%	63.000000	64.000000	64.000000
max	94.000000	93.000000	91.000000

```

from pandas_profiling import ProfileReport

df_Subset_random = pd.DataFrame(np.random.rand(10, 5), columns = ['a',
'b', 'c', 'd', 'e'])

profile = ProfileReport(df_subset, title = 'FIFA Profiling Report',
html = {'style': {'full_width': True}})

profile

{"version_major":2,"version_minor":0,"model_id":"9600f40753f248d58685f74738cb01ff"}

{"version_major":2,"version_minor":0,"model_id":"e735f1b0444343fb85dfbcd31ff0c757"}

{"version_major":2,"version_minor":0,"model_id":"c2d01c5c414b42419720f3b131898377"}

<IPython.core.display.HTML object>

```

Feature Analysis

- **Name:** 94% of names are distinct, J.Rodriguez is most common name
- **Player Position:** Corner Back (CB) and Goal Keeper (GK) are two highest number of positions
- **Overall:** Highest overall rating in the dataset is 93, and lowest is 47
- **Potential:** Highest potential in the dataset is 95, and lowest is 49
- **Wage:** Average wage in the dataset is 9017 EUR, while highest is 350000 EUR and lowest is 500 EUR
- **Age:** Oldest player in the dataset is 54, and the youngest player is 16
- **Club Name:** The clubs with most number of players in the dataset are PSG, Arsenal and Brentford.
- **Nationality:** Most players are from England, Germany, and Spain
- **Shooting:** Average shooting score is 52, while maximum score is 94 and minimum score is 18
- **Passing:** Average passing score is 57, while maximum score is 93 and minimum score is 25
- **Defending:** Average shooting score is 51.7, while maximum score is 91 and minimum score is 14

Feature Correlation Analysis

- **High positive correlation between Wage and Overall score (eg. players like Lewandowski and Messi)**
- **High positive correlation between Shooting and Passing**

- **High negative correlation between shooting and defending (eg. Strikers are not good defenders)**
- **Low correlation between Wage and Age (eg. younger players can get paid just as much as older players if they are good. Earling Holland of Manchester City FC is only 22 year old and is currently the highest wage earner in EPL)**
- **High positive correlation between Passing and Overall Score (eg. good passing skill is one of most important attribute in soccer, as it is a team sport)**

4. Goalkeeper Dataset Analysis

*For our model analysis, for simplicity and clarity, we decided to use only **goalkeeper (GK)** dataset. This allows us to:*

- Control attribute weight distribution (eg. Making sure no one weight overweights other weights by a lot)
- Improve Silhouette score (score that measures distance matrix between datapoint in a cluster)
- Easier web app deployment (narrow down attributes from 110)

```
df_gk=df[df.player_positions == "GK"]
df_gk.shape
```

```
(2132, 110)
```

```
df_gk.head()
```

```

    sofifa_id                                player_url \
5      200389  https://sofifa.com/player/200389/jan-oblak/220002
7      167495  https://sofifa.com/player/167495/manuel-neuer/...
8      192448  https://sofifa.com/player/192448/marc-andre-te...
12     192119  https://sofifa.com/player/192119/thibaut-court...
18     210257  https://sofifa.com/player/210257/ederson-santa...
```

	short_name	long_name	player_positions	overall
5	J. Oblak	Jan Oblak	GK	91
7	M. Neuer	Manuel Peter Neuer	GK	90
8	M. ter Stegen	Marc-André ter Stegen	GK	90
12	T. Courtois	Thibaut Courtois	GK	89
18	Ederson	Ederson Santana de Moraes	GK	89

	potential	value_eur	wage_eur	age	dob	height_cm
weight_kg \						
5	93	112000000.0	130000.0	28	1993-01-07	188

87						
7	90	13500000.0	86000.0	35	1986-03-27	193
93						
8	92	99000000.0	250000.0	29	1992-04-30	187
85						
12	91	85500000.0	250000.0	29	1992-05-11	199
96						
18	91	94000000.0	200000.0	27	1993-08-17	188
86						

	club_team_id	club_name	league_name
league_level \			
5	240.0	Atlético de Madrid	Spain Primera Division
1.0			
7	21.0	FC Bayern München	German 1. Bundesliga
1.0			
8	241.0	FC Barcelona	Spain Primera Division
1.0			
12	243.0	Real Madrid CF	Spain Primera Division
1.0			
18	10.0	Manchester City	English Premier League
1.0			

	club_position	club_jersey_number	club_loaned_from	club_joined \
5	GK	13.0	NaN	2014-07-16
7	GK	1.0	NaN	2011-07-01
8	GK	1.0	NaN	2014-07-01
12	GK	1.0	NaN	2018-08-09
18	GK	31.0	NaN	2017-07-01

	club_contract_valid_until	nationality_id	nationality_name \
5	2023.0	44	Slovenia
7	2023.0	21	Germany
8	2025.0	21	Germany
12	2026.0	7	Belgium
18	2026.0	54	Brazil

	nation_team_id	nation_position	nation_jersey_number
preferred_foot \			
5	NaN	NaN	NaN
Right			
7	1337.0	GK	1.0
Right			
8	NaN	NaN	NaN
Right			
12	1325.0	GK	1.0
Left			
18	NaN	NaN	NaN
Left			

	weak_foot	skill_moves	international_reputation	work_rate
body_type \				
5	3	1	5	Medium/Medium
Unique				
7	4	1	5	Medium/Medium
Unique				
8	4	1	4	Medium/Medium
Unique				
12	3	1	4	Medium/Medium
Unique				
18	3	1	3	Medium/Medium
Unique				

	real_face	release_clause_eur	player_tags \
5	Yes	238000000.0	NaN
7	Yes	223000000.0	NaN
8	Yes	210400000.0	NaN
12	Yes	181700000.0	NaN
18	Yes	181000000.0	NaN

	player_traits	pace	shooting
\			
5	GK Long Throw, Comes For Crosses	NaN	NaN
7	Leadership, GK Long Throw, Rushes Out Of Goal,...	NaN	NaN
8	Rushes Out Of Goal, Comes For Crosses, Saves w...	NaN	NaN
12	GK Long Throw, Comes For Crosses	NaN	NaN
18	Long Passer (AI), Rushes Out Of Goal, Comes Fo...	NaN	NaN

	passing	dribbling	defending	physic	attacking_crossing \
5	NaN	NaN	NaN	NaN	13
7	NaN	NaN	NaN	NaN	15
8	NaN	NaN	NaN	NaN	18
12	NaN	NaN	NaN	NaN	14
18	NaN	NaN	NaN	NaN	20

	attacking_finishing	attacking_heading_accuracy
attacking_short_passing \		
5	11	15
43		
7	13	25
60		
8	14	11
61		
12	14	13

33
18
61

14

14

\	attacking_volleys	skill_dribbling	skill_curve	skill_fk_accuracy
5	13	12	13	14
7	11	30	14	11
8	14	21	18	12
12	12	13	19	20
18	18	23	15	20

	skill_long_passing	skill_ball_control	movement_acceleration	\
5	40	30	43	
7	68	46	54	
8	63	30	38	
12	35	23	42	
18	66	40	64	

	movement_sprint_speed	movement_agility	movement_reactions	\
5	60	67	88	
7	60	51	87	
8	50	39	86	
12	52	62	84	
18	63	60	88	

	movement_balance	power_shot_power	power_jumping	
power_stamina	\			
5	49	59	78	41
7	35	68	77	43
8	43	66	79	35
12	45	56	68	38
18	48	70	66	41

	power_strength	power_long_shots	mentality_aggression	\
5	78	12	34	
7	80	16	29	
8	78	10	43	
12	70	17	23	

18	78	18	38
mentality_interceptions		mentality_positioning	
mentality_vision \			
5	19	11	65
7	30	12	70
8	22	11	70
12	15	13	44
18	27	20	70

mentality_penalties	mentality_composure
defending_marking_awareness \	
5	11
27	68
7	47
17	70
8	25
25	70
12	27
20	66
18	51
29	70

defending_standing_tackle	defending_sliding_tackle
goalkeeping_diving \	
5	12
87	18
7	10
88	11
8	13
88	10
12	18
84	16
18	15
87	8

goalkeeping_handling	goalkeeping_kicking	goalkeeping_positioning
\		
5	92	78
		90
7	88	91
		89
8	85	88
		88

12	89	74	86
18	82	93	88

	goalkeeping_reflexes	goalkeeping_speed	ls	st	rs	lw	lf
cf rf \							
5	90	50.0	33+3	33+3	33+3	32	35
35 35							
7	88	56.0	40+3	40+3	40+3	40	43
43 43							
8	90	43.0	35+3	35+3	35+3	35	38
38 38							
12	88	46.0	31+3	31+3	31+3	29	31
31 31							
18	88	64.0	40+3	40+3	40+3	41	43
43 43							

	rw	lam	cam	ram	lm	lcm	cm	rcm	rm	lwb	ldm
cdm \											
5	32	38+3	38+3	38+3	35+3	38+3	38+3	38+3	35+3	32+3	36+3
36+3											
7	40	47+3	47+3	47+3	44+3	50+3	50+3	50+3	44+3	37+3	43+3
43+3											
8	35	42+3	42+3	42+3	39+3	45+3	45+3	45+3	39+3	33+3	41+3
41+3											
12	29	32+3	32+3	32+3	31+3	32+3	32+3	32+3	31+3	29+3	31+3
31+3											
18	41	47+3	47+3	47+3	44+3	49+3	49+3	49+3	44+3	37+3	44+3
44+3											

	rdm	rwb	lb	lcb	cb	rcb	rb	gk	\
5	36+3	32+3	32+3	33+3	33+3	33+3	32+3	89+3	
7	43+3	37+3	35+3	34+3	34+3	34+3	35+3	88+2	
8	41+3	33+3	31+3	33+3	33+3	33+3	31+3	88+3	
12	31+3	29+3	29+3	29+3	29+3	29+3	29+3	86+3	
18	44+3	37+3	36+3	35+3	35+3	35+3	36+3	87+3	

	player_face_url	\
5	https://cdn.sofifa.net/players/200/389/22_120.png	
7	https://cdn.sofifa.net/players/167/495/22_120.png	
8	https://cdn.sofifa.net/players/192/448/22_120.png	
12	https://cdn.sofifa.net/players/192/119/22_120.png	
18	https://cdn.sofifa.net/players/210/257/22_120.png	

	club_logo_url	\
5	https://cdn.sofifa.net/teams/240/60.png	
7	https://cdn.sofifa.net/teams/21/60.png	
8	https://cdn.sofifa.net/teams/241/60.png	

```

12 https://cdn.sofifa.net/teams/243/60.png
18 https://cdn.sofifa.net/teams/10/60.png

club_flag_url \
5 https://cdn.sofifa.net/flags/es.png
7 https://cdn.sofifa.net/flags/de.png
8 https://cdn.sofifa.net/flags/es.png
12 https://cdn.sofifa.net/flags/es.png
18 https://cdn.sofifa.net/flags/gb-eng.png

nation_logo_url \
5 NaN
7 https://cdn.sofifa.net/teams/1337/60.png
8 NaN
12 https://cdn.sofifa.net/teams/1325/60.png
18 NaN

nation_flag_url
5 https://cdn.sofifa.net/flags/si.png
7 https://cdn.sofifa.net/flags/de.png
8 https://cdn.sofifa.net/flags/de.png
12 https://cdn.sofifa.net/flags/be.png
18 https://cdn.sofifa.net/flags/br.png

```

```

df_gk = df_gk.rename(columns={'short_name':
'goalkeeping_diving': 'diving', 'goalkeeping_handling': 'handling'
, 'goalkeeping_kicking': 'kicking', 'goalkeeping_positioning': 'positionin
g', 'goalkeeping_reflexes': 'reflexes', 'goalkeeping_speed': 'speed'})

```

```

attributes= ['name', 'overall', 'age', 'wage_eur', 'value_eur', 'diving',
'handling', 'kicking', 'positioning', 'reflexes', 'speed']

```

We handpicked attributes that gives us a high Silhouette score. The process of choosing the attribute was done through trial-and-error, as we experimented with different combination of attributes to obtain high Silhouette score, as well avoid overweight to any one attribute that will vastly influence the model's performance.

Attributes we used for our models are as follows:

- Name
- Overall (score)
- Age
- Wage (in Euros)
- Value (in Euros)
- Diving
- Handling
- Kicking
- Positioning
- Reflexes

- Speed

```
df_gk_final = df_gk[attributes].copy()
df_gk.reset_index(drop=True, inplace=True)
df_gk.head()
```

	sofifa_id	player_url \
0	200389	https://sofifa.com/player/200389/jan-oblak/220002
1	167495	https://sofifa.com/player/167495/manuel-neuer/...
2	192448	https://sofifa.com/player/192448/marc-andre-te...
3	192119	https://sofifa.com/player/192119/thibaut-court...
4	210257	https://sofifa.com/player/210257/ederson-santa...

	name	long_name	player_positions	overall
0	J. Oblak	Jan Oblak	GK	91
1	M. Neuer	Manuel Peter Neuer	GK	90
2	M. ter Stegen	Marc-André ter Stegen	GK	90
3	T. Courtois	Thibaut Courtois	GK	89
4	Ederson	Ederson Santana de Moraes	GK	89

	potential	value_eur	wage_eur	age	dob	height_cm
0	93	112000000.0	130000.0	28	1993-01-07	188
1	90	13500000.0	86000.0	35	1986-03-27	193
2	92	99000000.0	250000.0	29	1992-04-30	187
3	91	85500000.0	250000.0	29	1992-05-11	199
4	91	94000000.0	200000.0	27	1993-08-17	188

	club_team_id	club_name	league_name
0	240.0	Atlético de Madrid	Spain Primera Division
1	21.0	FC Bayern München	German 1. Bundesliga
2	241.0	FC Barcelona	Spain Primera Division
3	243.0	Real Madrid CF	Spain Primera Division
4	10.0	Manchester City	English Premier League

1.0

	club_position	club_jersey_number	club_loaned_from	club_joined	\
0	GK	13.0	NaN	2014-07-16	
1	GK	1.0	NaN	2011-07-01	
2	GK	1.0	NaN	2014-07-01	
3	GK	1.0	NaN	2018-08-09	
4	GK	31.0	NaN	2017-07-01	

	club_contract_valid_until	nationality_id	nationality_name
nation_team_id \			
0	2023.0	44	Slovenia
NaN			
1	2023.0	21	Germany
1337.0			
2	2025.0	21	Germany
NaN			
3	2026.0	7	Belgium
1325.0			
4	2026.0	54	Brazil
NaN			

	nation_position	nation_jersey_number	preferred_foot	weak_foot	\
0	NaN	NaN	Right	3	
1	GK	1.0	Right	4	
2	NaN	NaN	Right	4	
3	GK	1.0	Left	3	
4	NaN	NaN	Left	3	

	skill_moves	international_reputation	work_rate	body_type
real_face \				
0	1	5	Medium/Medium	Unique
Yes				
1	1	5	Medium/Medium	Unique
Yes				
2	1	4	Medium/Medium	Unique
Yes				
3	1	4	Medium/Medium	Unique
Yes				
4	1	3	Medium/Medium	Unique
Yes				

	release_clause_eur	player_tags	\
0	238000000.0	NaN	
1	223000000.0	NaN	
2	210400000.0	NaN	
3	181700000.0	NaN	
4	181000000.0	NaN	

player_traits pace shooting

passing \			
0	GK Long Throw, Comes For Crosses	NaN	NaN
NaN			
1	Leadership, GK Long Throw, Rushes Out Of Goal,...	NaN	NaN
NaN			
2	Rushes Out Of Goal, Comes For Crosses, Saves w...	NaN	NaN
NaN			
3	GK Long Throw, Comes For Crosses	NaN	NaN
NaN			
4	Long Passer (AI), Rushes Out Of Goal, Comes Fo...	NaN	NaN
NaN			

	dribbling	defending	physic	attacking_crossing
attacking_finishing \				
0	NaN	NaN	NaN	13
11				
1	NaN	NaN	NaN	15
13				
2	NaN	NaN	NaN	18
14				
3	NaN	NaN	NaN	14
14				
4	NaN	NaN	NaN	20
14				

	attacking_heading_accuracy	attacking_short_passing
attacking_volleys \		
0	15	43
13		
1	25	60
11		
2	11	61
14		
3	13	33
12		
4	14	61
18		

	skill_dribbling	skill_curve	skill_fk_accuracy	skill_long_passing
\				
0	12	13	14	40
1	30	14	11	68
2	21	18	12	63
3	13	19	20	35
4	23	15	20	66

	skill_ball_control	movement_acceleration	movement_sprint_speed	\
0	30	43	60	
1	46	54	60	
2	30	38	50	
3	23	42	52	
4	40	64	63	

	movement_agility	movement_reactions	movement_balance
power_shot_power \			
0	67	88	49
59			
1	51	87	35
68			
2	39	86	43
66			
3	62	84	45
56			
4	60	88	48
70			

	power_jumping	power_stamina	power_strength	power_long_shots	\
0	78	41	78	12	
1	77	43	80	16	
2	79	35	78	10	
3	68	38	70	17	
4	66	41	78	18	

	mentality_aggression	mentality_interceptions
mentality_positioning \		
0	34	19
11		
1	29	30
12		
2	43	22
11		
3	23	15
13		
4	38	27
20		

	mentality_vision	mentality_penalties	mentality_composure	\
0	65	11	68	
1	70	47	70	
2	70	25	70	
3	44	27	66	
4	70	51	70	

	defending_marking_awareness	defending_standing_tackle \
0	27	12
1	17	10
2	25	13
3	20	18
4	29	15

	defending_sliding_tackle	diving	handling	kicking	positioning
0	18	87	92	78	90
1	11	88	88	91	89
2	10	88	85	88	88
3	16	84	89	74	86
4	8	87	82	93	88

	speed	ls	st	rs	lw	lf	cf	rf	rw	lam	cam	ram	lm
0	50.0	33+3	33+3	33+3	32	35	35	35	32	38+3	38+3	38+3	35+3
1	56.0	40+3	40+3	40+3	40	43	43	43	40	47+3	47+3	47+3	44+3
2	43.0	35+3	35+3	35+3	35	38	38	38	35	42+3	42+3	42+3	39+3
3	46.0	31+3	31+3	31+3	29	31	31	31	29	32+3	32+3	32+3	31+3
4	64.0	40+3	40+3	40+3	41	43	43	43	41	47+3	47+3	47+3	44+3

	cm	rcm	rm	lwb	ldm	cdm	rdm	rwb	lb	lcb	cb
0	38+3	38+3	35+3	32+3	36+3	36+3	36+3	32+3	32+3	33+3	33+3
1	50+3	50+3	44+3	37+3	43+3	43+3	43+3	37+3	35+3	34+3	34+3
2	45+3	45+3	39+3	33+3	41+3	41+3	41+3	33+3	31+3	33+3	33+3
3	32+3	32+3	31+3	29+3	31+3	31+3	31+3	29+3	29+3	29+3	29+3
4	49+3	49+3	44+3	37+3	44+3	44+3	44+3	37+3	36+3	35+3	35+3

	rb	gk	player_face_url \
0	32+3	89+3	https://cdn.sofifa.net/players/200/389/22_120.png
1	35+3	88+2	https://cdn.sofifa.net/players/167/495/22_120.png
2	31+3	88+3	https://cdn.sofifa.net/players/192/448/22_120.png

```
3 29+3 86+3 https://cdn.sofifa.net/players/192/119/22_120.png
4 36+3 87+3 https://cdn.sofifa.net/players/210/257/22_120.png
```

```
club_logo_url \
0 https://cdn.sofifa.net/teams/240/60.png
1 https://cdn.sofifa.net/teams/21/60.png
2 https://cdn.sofifa.net/teams/241/60.png
3 https://cdn.sofifa.net/teams/243/60.png
4 https://cdn.sofifa.net/teams/10/60.png
```

```
club_flag_url \
0 https://cdn.sofifa.net/flags/es.png
1 https://cdn.sofifa.net/flags/de.png
2 https://cdn.sofifa.net/flags/es.png
3 https://cdn.sofifa.net/flags/es.png
4 https://cdn.sofifa.net/flags/gb-eng.png
```

```
nation_logo_url \
0 NaN
1 https://cdn.sofifa.net/teams/1337/60.png
2 NaN
3 https://cdn.sofifa.net/teams/1325/60.png
4 NaN
```

```
nation_flag_url
0 https://cdn.sofifa.net/flags/si.png
1 https://cdn.sofifa.net/flags/de.png
2 https://cdn.sofifa.net/flags/de.png
3 https://cdn.sofifa.net/flags/be.png
4 https://cdn.sofifa.net/flags/br.png
```



Fun fact: The goalkeeper with the highest overall rating in the FIFA22 dataset is Jan Oblak, who plays for Atlético Madrid.

```
df_gk_final.shape
```

```
(2132, 11)
```

Finally, our model will use Goalkeeper dataset that has 2124 distinct players and 11 attributes.

```
profile_gk = ProfileReport(df_gk_final, title = 'Goalkeepers Profile Report', html = {'style': {'full_width': True}})
```

```
profile_gk
```

```
{"version_major":2,"version_minor":0,"model_id":"ee4e667974d3423ebec41albe22d562a"}
```

```
{"version_major":2,"version_minor":0,"model_id":"af562caf5d3647678a91bfadafb0eea6"}
```

```
{"version_major":2,"version_minor":0,"model_id":"325d4aabd4f24261a847c4bb56d2afe5"}
```

```
<IPython.core.display.HTML object>
```

Feature Analysis

- **Name:** 99% of names are distinct

- **Overall:** Highest overall rating in the dataset is 91, and lowest is 47, with the mean rating of 64
- **Age:** Oldest goalkeeper in the dataset is 43, and youngest is 16. Average age of goalkeeper is 26
- **Wage:** Average wage in the dataset is 6349 EUR, while highest is 250000 EUR and lowest is 500 EUR
- **Value:** Average value of goalkeeper in the dataset is 1,930,764 EUR, while highest is 119,500,000 EUR and lowest is 9,000 EUR
- **Diving:** Average diving score is 65, while maximum score is 91 and minimum score is 41
- **Handling:** Average Handling score is 63, while maximum score is 92 and minimum score is 41
- **Kicking:** Average Kicking score is 62, while maximum score is 93 and minimum score is 39
- **Positioning:** Average Positioning score is 63, while maximum score is 92 and minimum score is 38
- **Reflexes:** Average Reflexes score is 66, while maximum score is 90 and minimum score is 35
- **Speed:** Average Speed score is 36, while maximum score is 65 and minimum score is 15

Feature Correlation Analysis

- **High positive correlation between Diving and Overall score:** Often, goalkeeper must dive from their stationary position to block the shot from the shooter. The goalkeeper that can dive better has higher chance of blocking the shot, therefore, gets higher score
- **High positive correlation between Diving and Reflexes:** The goalkeeper must make a decision instantly before they dive left or dive right to block the shot. The goalkeeper with faster reflexes with therefore, dive better.
- **No correlation between Age and Value:** Soccer player's value comes from their performance in games, and goalkeepers at age that plays well in games will earn high valuation.
- **High correlation between Handling and Positioning:** Goalkeepers that handles balls well are usually better positioned to receive the ball.

5. Model Analysis

Professional Soccer clubs has to invest a lot of resources, time and money in the recruitment of best players available. The use of clustering on the database of football players based on their performance data is useful for prototyping potentially successful players, and also for providing insights to football managers and scouts when assessing players.

The clustering model techniques allows us to use different types of variables (strings, numerical, categorical) and different types of attributes to be taken into account. The weight is objectively assigned to the distance matrix associated to each set of attributes during the optimization process (D'Urso, 2022).

The weights show how relevant each attribute type is in the clustering output. The benefit of clustering is that it is robust against outliers. The obtained weights allow to understand which is the most relevant set of attributes in partitioning the players into clustering groups (D'Urso, 2022).

```
from pycaret.clustering import *
```

```
exp_clu101 = setup(data=df_gk_final, transformation=True, normalize =
True, normalize_method = 'robust', ignore_features =
['name'], session_id = 123)
```

	Description	Value
0	session_id	123
1	Original Data	(2132, 11)
2	Missing Values	True
3	Numeric Features	10
4	Categorical Features	0
5	Ordinal Features	False
6	High Cardinality Features	False
7	High Cardinality Method	None
8	Transformed Data	(2132, 10)
9	CPU Jobs	-1
10	Use GPU	False
11	Log Experiment	False
12	Experiment Name	cluster-default-name
13	USI	d54d
14	Imputation Type	simple
15	Iterative Imputation Iteration	None
16	Numeric Imputer	mean
17	Iterative Imputation Numeric Model	None
18	Categorical Imputer	mode
19	Iterative Imputation Categorical Model	None
20	Unknown Categoricals Handling	least_frequent
21	Normalize	True
22	Normalize Method	robust
23	Transformation	True
24	Transformation Method	yeo-johnson
25	PCA	False
26	PCA Method	None
27	PCA Components	None
28	Ignore Low Variance	False
29	Combine Rare Levels	False
30	Rare Level Threshold	None
31	Numeric Binning	False
32	Remove Outliers	False

33	Outliers Threshold	None
34	Remove Multicollinearity	False
35	Multicollinearity Threshold	None
36	Remove Perfect Collinearity	False
37	Clustering	False
38	Clustering Iteration	None
39	Polynomial Features	False
40	Polynomial Degree	None
41	Trigonometry Features	False
42	Polynomial Threshold	None
43	Group Features	False
44	Feature Selection	False
45	Feature Selection Method	classic
46	Features Selection Threshold	None
47	Feature Interaction	False
48	Feature Ratio	False
49	Interaction Threshold	None

```

INFO:logs:create_model_container: 0
INFO:logs:master_model_container: 0
INFO:logs:display_container: 1
INFO:logs:Pipeline(memory=None,
    steps=[('dtypes',
            DataTypes_Auto_infer(categorical_features=[],
                                display_types=True,
                                features_todrop=['name'],
                                id_columns=[],
                                ml_usecase='regression',
                                numerical_features=[],
                                target='UNSUPERVISED_DUMMY_TARGET',
                                time_features=[])),
            ('imputer',
             Simple_Imputer(categorical_strategy='most frequent',
                             fill_value_categorical=None,
                             fill_value_numerica...
                             target='UNSUPERVISED_DUMMY_TARGET')),
            ('binn', 'passthrough'), ('rem_outliers',
            'passthrough'),
            ('cluster_all', 'passthrough'),
            ('dummy',
             Dummify(target='UNSUPERVISED_DUMMY_TARGET')),
            ('fix_perfect', 'passthrough'),
            ('clean_names', Clean_Colum_Names()),
            ('feature_select', 'passthrough'), ('fix_multi',
            'passthrough'),
            ('dfs', 'passthrough'), ('pca', 'passthrough')],
    verbose=False)

```

```
INFO:logs:setup() succesfully
completed.....
```

- We first setup the Pycaret cluster models environment
- In the steup, we will normalize the data using yeo-johnson transformation and then scale it using Robust Scaler Technique
- The above steps have been underataken to reduce the impact of outliers present in "wages_eur" and "value_eur" variables.

```
models()
```

```
INFO:logs:gpu_param set to False
```

ID	Name \	Reference
kmeans	K-Means Clustering	sklearn.cluster._kmeans.KMeans
ap	Affinity Propagation	sklearn.cluster._affinity_propagation.Affinity...
meanshift	Mean Shift Clustering	sklearn.cluster._mean_shift.MeanShift
sc	Spectral Clustering	sklearn.cluster._spectral.SpectralClustering
hclust	Agglomerative Clustering	sklearn.cluster._agglomerative.AgglomerativeCl...
dbscan	Density-Based Spatial Clustering	sklearn.cluster._dbscan.DBSCAN
optics	OPTICS Clustering	sklearn.cluster._optics.OPTICS
birch	Birch Clustering	sklearn.cluster._birch.Birch
kmodes	K-Modes Clustering	kmodes.kmodes.KModes

- The different clustering algorithms available in Pycaret is as shown above
- We will explore K-Means, DBSCAN and K-Modes, before finalizing the best model amongst them

```
kmeans = create_model('kmeans')
```

Silhouette Index \	Calinski-Harabasz	Davies-Bouldin	Homogeneity	Rand
0	0.2277	1331.0122	1.4033	0
0				
Completeness				
0	0			

```
INFO:logs:create_model_container: 1
INFO:logs:master_model_container: 1
INFO:logs:display_container: 2
INFO:logs:KMeans(algorithm='auto', copy_x=True, init='k-means++',
max_iter=300,
    n_clusters=4, n_init=10, n_jobs=-1,
precompute_distances='deprecated',
    random_state=123, tol=0.0001, verbose=0)
INFO:logs:create_model() succesfully
completed.....
```

```
kmodes = create_model('kmodes')
```

	Silhouette	Calinski-Harabasz	Davies-Bouldin	Homogeneity	Rand
Index \					
0	-0.1224	33.3701	11.4024		0
0					

	Completeness
0	0

```
INFO:logs:create_model_container: 2
INFO:logs:master_model_container: 2
INFO:logs:display_container: 3
INFO:logs:KModes(cat_dissim=<function matching_dissim at
0x7fe2148873a0>, init='Cao',
    max_iter=100, n_clusters=4, n_init=1, n_jobs=-1,
random_state=123,
    verbose=0)
INFO:logs:create_model() succesfully
completed.....
```

```
dbscan=create_model('dbscan')
```

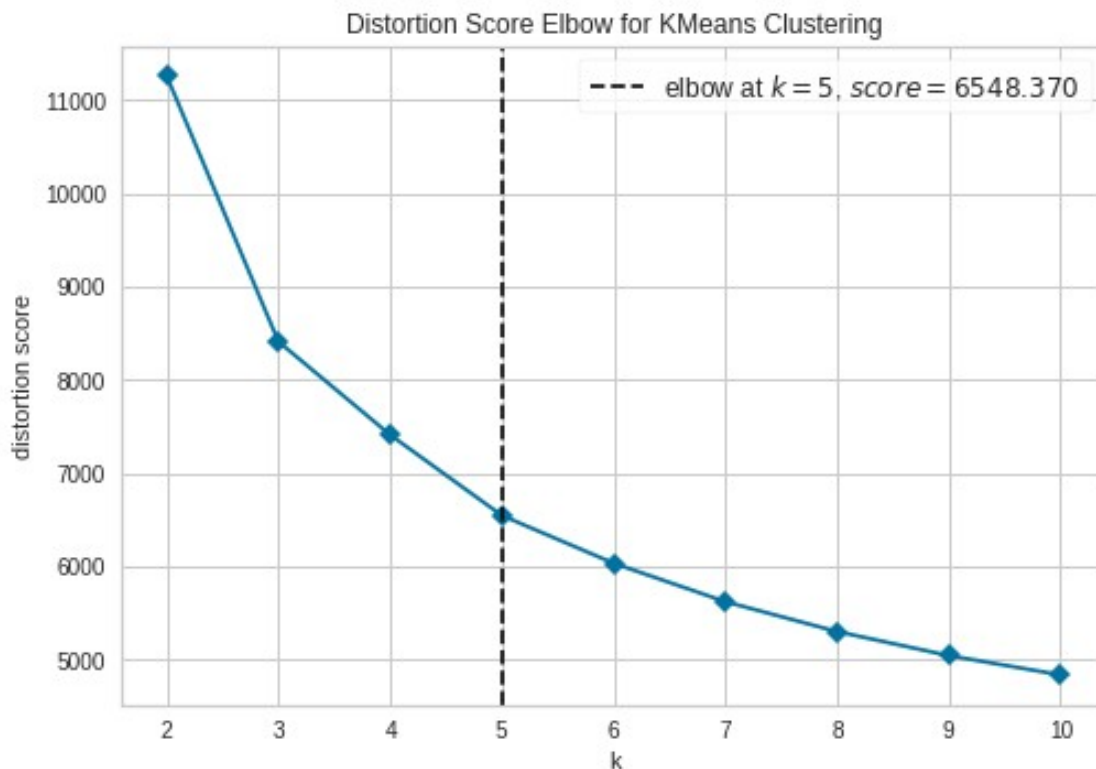
	Silhouette	Calinski-Harabasz	Davies-Bouldin	Homogeneity	Rand
Index \					
0	-0.2707	0.4543	3.3775		0
0					

	Completeness
0	0

```
INFO:logs:create_model_container: 3
INFO:logs:master_model_container: 3
INFO:logs:display_container: 4
INFO:logs:DBSCAN(algorithm='auto', eps=0.5, leaf_size=30,
metric='euclidean',
    metric_params=None, min_samples=5, n_jobs=-1, p=None)
INFO:logs:create_model() succesfully
completed.....
```

- After exploring different models, we can conclude that K-means technique is best suited for this dataset

```
plot_model(kmeans, plot='elbow')
```



```
INFO:logs:Visual Rendered Successfully
INFO:logs:plot_model() succesfully
completed.....
```

#Elbow method: In this method, we will change the number of clusters (K) between 1 and 10, and for each value of K, we will calculate the Within-Cluster Sum of Square (WCSS). The WCSS is the sum of squared distance between each point and the centroid in a cluster. When we we plot the WCSS with a paticluar K value, the plot looks like an Elbow. As the number of clusters increases, the WCSS value will start to decrease. So, WCSS value is largest when K = 1, and the plot will rapidly change at a point and thus creating an elbow shape. From this point, the graph starts to move almost parallel to the X-axis. The K value corresponding to this point is the optimal K value or an optimal number of clusters.

- For our dataset, K = 5, which means beyond 5 clusters, the efficiency of the model will not increase.

```
kmeans5 = create_model('kmeans', num_clusters=5)
```

	Silhouette	Calinski-Harabasz	Davies-Bouldin	Homogeneity	Rand
Index \					
0	0.2308	1199.5065	1.3362	0	
0					

```

    Completeness
0          0

INFO:logs:create_model_container: 4
INFO:logs:master_model_container: 4
INFO:logs:display_container: 5
INFO:logs:KMeans(algorithm='auto', copy_x=True, init='k-means++',
max_iter=300,
            n_clusters=5, n_init=10, n_jobs=-1,
precompute_distances='deprecated',
            random_state=123, tol=0.0001, verbose=0)
INFO:logs:create_model() succesfully
completed.....

```

- When we model for K=5 clusters, we get a silhouette score of 0.2308
- The *silhouette value* is a measure of how similar an object is to its own cluster (cohesion) compared to other clusters (separation). The silhouette ranges from -1 to +1, where a high value indicates that the object is well matched to its own cluster and poorly matched to neighboring clusters.
- Although not very high, this value seems the highest we can obtain from this dataset.
- The model also has a Calinski-Harabasz score of 1199.5065. The Calinski-Harabasz index is the ratio of the sum of between-clusters dispersion and of inter-cluster dispersion for all clusters. The higher the score, the better the performance.
- In general, the model does show separation but lacks cohesion.

```

plot_model(kmeans5)

INFO:logs:Rendering Visual

INFO:logs:Visual Rendered Successfully
INFO:logs:plot_model() succesfully
completed.....

```

Plot shows the five clusters after running the K-Means model

Clusters are formed based on player value

- Cluster 2 is the cluster with all top players, followed by cluster 3, Cluster 4, Cluster 0 and Cluster 1

```

plot_model(kmeans5, plot = 'tsne')

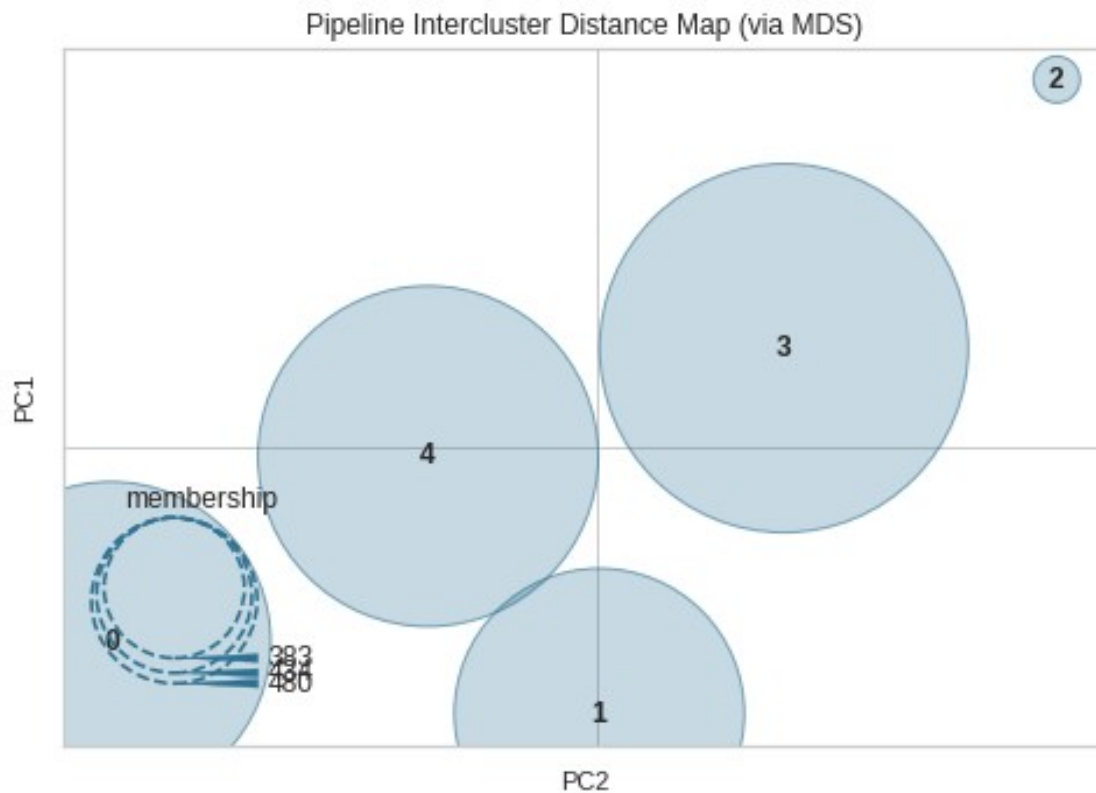
INFO:logs:Rendering Visual

INFO:logs:Visual Rendered Successfully
INFO:logs:plot_model() succesfully
completed.....

```

This is a t-distributed stochastic neighbor embedding plot which is a three-dimensional view of the clusters

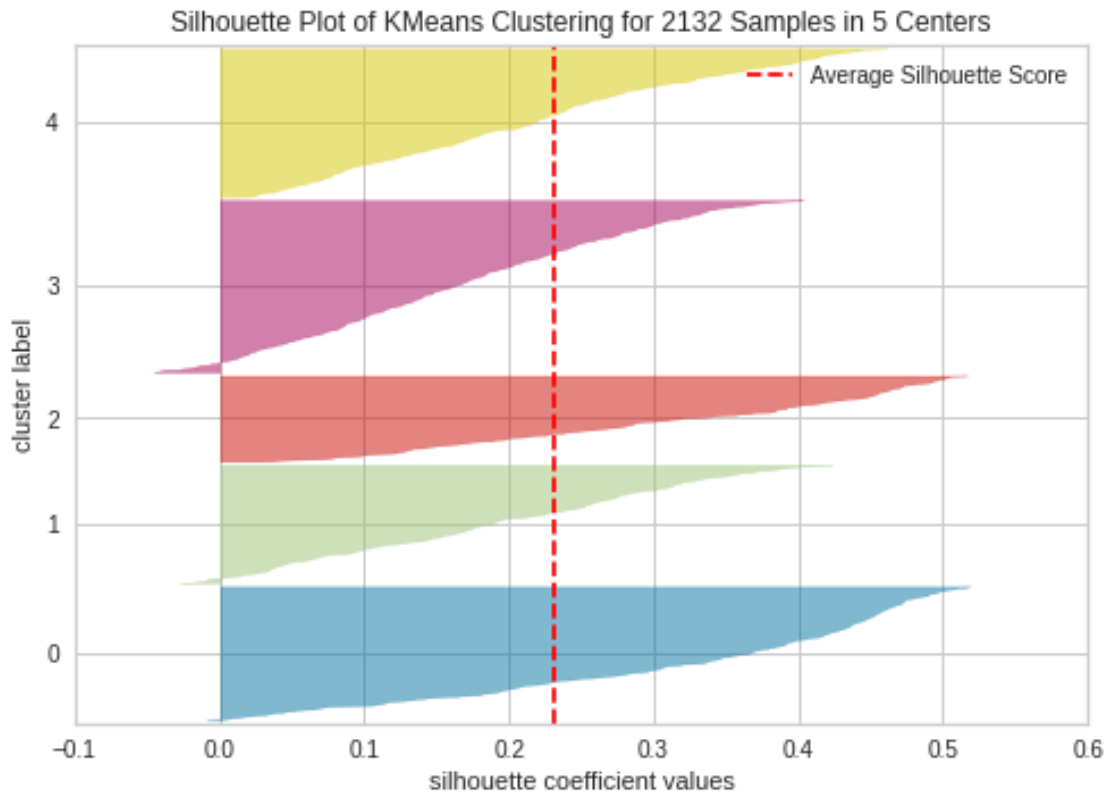
```
plot_model(kmeans5, plot = 'distance')
```



```
INFO:logs:Visual Rendered Successfully
INFO:logs:plot_model() succesfully
completed.....
```

Intercluster distance maps show an embedding of the cluster centers in two dimensions with the distance to other centers preserved

```
plot_model(kmeans5, plot = 'silhouette')
```



```
INFO:logs:Visual Rendered Successfully
INFO:logs:plot_model() successfully
completed.....
```

The silhouette plot displays a measure of how close each point in one cluster is to points in the neighboring clusters and thus provides a way to assess parameters like number of clusters visually

```
plot_model(kmeans5, plot = 'distribution') #to see size of clusters
```

```
INFO:logs:Rendering Visual
INFO:logs:Visual Rendered Successfully
INFO:logs:plot_model() successfully
completed.....
```

Plot shows the size of each cluster

```
plot_model(kmeans, plot = 'distribution', feature = 'overall')
```

```
INFO:logs:Rendering Visual
INFO:logs:Visual Rendered Successfully
INFO:logs:plot_model() successfully
completed.....
```

Plot shows distribution of attribute "overall" across clusters

```
plot_model(kmeans, plot = 'distribution', feature = 'age')
```

```
INFO:logs:Rendering Visual
```

```
INFO:logs:Visual Rendered Successfully
```

```
INFO:logs:plot_model() successfully
```

```
completed.....
```

```
Plot shows distribution of attribute "age" across clusters
```

```
plot_model(kmeans, plot = 'distribution', feature = 'wage_eur')
```

```
INFO:logs:Rendering Visual
```

```
INFO:logs:Visual Rendered Successfully
```

```
INFO:logs:plot_model() successfully
```

```
completed.....
```

```
Plot shows distribution of attribute "wage_eur" across clusters
```

```
plot_model(kmeans, plot = 'distribution', feature = 'value_eur')
```

```
INFO:logs:Rendering Visual
```

```
INFO:logs:Visual Rendered Successfully
```

```
INFO:logs:plot_model() successfully
```

```
completed.....
```

```
Plot shows distribution of attribute "value_eur" across clusters
```

```
plot_model(kmeans, plot = 'distribution', feature = 'diving')
```

```
INFO:logs:Rendering Visual
```

```
INFO:logs:Visual Rendered Successfully
```

```
INFO:logs:plot_model() successfully
```

```
completed.....
```

```
Plot shows distribution of attribute "diving" across clusters
```

```
plot_model(kmeans, plot = 'distribution', feature = 'handling')
```

```
INFO:logs:Rendering Visual
```

```
INFO:logs:Visual Rendered Successfully
```

```
INFO:logs:plot_model() successfully
```

```
completed.....
```

```
Plot shows distribution of attribute "handling" across clusters
```

```
plot_model(kmeans, plot = 'distribution', feature = 'kicking')
```

```
INFO:logs:Rendering Visual
```

```
INFO:logs:Visual Rendered Successfully
INFO:logs:plot_model() succesfully
completed.....
```

Plot shows distribution of attribute "kicking" across clusters

```
plot_model(kmeans, plot = 'distribution', feature = 'positioning')
```

```
INFO:logs:Rendering Visual
```

```
INFO:logs:Visual Rendered Successfully
INFO:logs:plot_model() succesfully
completed.....
```

Plot shows distribution of attribute "positioning" across clusters

```
plot_model(kmeans, plot = 'distribution', feature = 'reflexes')
```

```
INFO:logs:Rendering Visual
```

```
INFO:logs:Visual Rendered Successfully
INFO:logs:plot_model() succesfully
completed.....
```

Plot shows distribution of attribute "reflexes" across clusters

```
plot_model(kmeans, plot = 'distribution', feature = 'speed')
```

```
INFO:logs:Rendering Visual
```

```
INFO:logs:Visual Rendered Successfully
INFO:logs:plot_model() succesfully
completed.....
```

Plot shows distribution of attribute "speed" across clusters

```
kmean_results = assign_model(kmeans5)
kmean_results.head()
```

```
INFO:logs:Initializing assign_model()
INFO:logs:assign_model(model=KMeans(algorithm='auto', copy_x=True,
init='k-means++', max_iter=300,
      n_clusters=5, n_init=10, n_jobs=-1,
precompute_distances='deprecated',
      random_state=123, tol=0.0001, verbose=0), transformation=False,
score=True, verbose=True)
INFO:logs:Checking exceptions
INFO:logs:Determining Trained Model
INFO:logs:Trained Model : K-Means Clustering
INFO:logs:Copying data
INFO:logs:(2132, 12)
INFO:logs:assign_model() succesfully
completed.....
```

	name	overall	age	wage_eur	value_eur	diving
handling \						
5	J. Oblak	91	28	130000.0	112000000.0	87
92						
7	M. Neuer	90	35	86000.0	13500000.0	88
88						
8	M. ter Stegen	90	29	250000.0	99000000.0	88
85						
12	T. Courtois	89	29	250000.0	85500000.0	84
89						
18	Ederson	89	27	200000.0	94000000.0	87
82						

	kicking	positioning	reflexes	speed	Cluster
5	78	90	90	50.0	Cluster 2
7	91	89	88	56.0	Cluster 2
8	88	88	90	43.0	Cluster 2
12	74	86	88	46.0	Cluster 2
18	93	88	88	64.0	Cluster 2

Gives the final output dataframe

```
save_model(kmeans, 'Final Kmeans Model CSML-Oct2022')
```

This is the final model which will be used for app deployment

```
df_kmeans = kmean_results.copy()
```

```
df_kmeans.head()
```

	name	overall	age	wage_eur	value_eur	diving
handling \						
5	J. Oblak	91	28	130000.0	112000000.0	87
92						
7	M. Neuer	90	35	86000.0	13500000.0	88
88						
8	M. ter Stegen	90	29	250000.0	99000000.0	88
85						
12	T. Courtois	89	29	250000.0	85500000.0	84
89						
18	Ederson	89	27	200000.0	94000000.0	87
82						

	kicking	positioning	reflexes	speed	Cluster
5	78	90	90	50.0	Cluster 2
7	91	89	88	56.0	Cluster 2
8	88	88	90	43.0	Cluster 2
12	74	86	88	46.0	Cluster 2
18	93	88	88	64.0	Cluster 2

```
df_kmeans_new = df_kmeans.replace(['Cluster 0', 'Cluster 1', 'Cluster 2', 'Cluster 3', 'Cluster 4'], [0, 1, 2, 3, 4])
df_kmeans_new
```

	name	overall	age	wage_eur	value_eur	diving
5	J. Oblak	91	28	130000.0	112000000.0	87
7	M. Neuer	90	35	86000.0	13500000.0	88
8	M. ter Stegen	90	29	250000.0	99000000.0	88
12	T. Courtois	89	29	250000.0	85500000.0	84
18	Ederson	89	27	200000.0	94000000.0	87
...	...	...	...	...	...	...
19210	Gao Xiang	48	20	800.0	60000.0	46
19216	H. Wiles-Richards	48	19	1000.0	110000.0	52
19225	D. Da Silva	47	21	500.0	90000.0	48
19227	A. Shaikh	47	18	500.0	110000.0	49
19231	R. By	47	22	500.0	90000.0	49

	handling	kicking	positioning	reflexes	speed	Cluster
5	92	78	90	90	50.0	2
7	88	91	89	88	56.0	2
8	85	88	88	90	43.0	2
12	89	74	86	88	46.0	2
18	82	93	88	88	64.0	2
...	...	...	...	...	...	...
19210	49	51	48	51	24.0	0
19216	49	49	43	48	33.0	0
19225	45	45	46	47	26.0	0
19227	41	39	45	49	19.0	0
19231	46	43	47	52	21.0	0

[2132 rows x 12 columns]

```
import plotly.express as px
fig = px.parallel_coordinates(df_kmeans_new, color="Cluster",
dimensions=['overall', 'age', 'wage_eur', 'value_eur', 'diving', 'handling',
'kicking', 'positioning', 'reflexes', 'speed'],
```

```
color_continuous_scale=px.colors.diverging.Tealrose)
fig.show()
```

The above plot is known as a parallel coordinate plot. Here each row in the output dataset is plotted as a line. Each attribute is represented by a point on the line. In this plot, we can see how the attributes are distributed across various clusters

6. Conclusion with Outcomes & Improvements

Our goal for this assignment was by using clustering on a database of players, we are able to cluster players based on attribute scores. One application of our model would be, generating a list of top players that the soccer scouts might be interested in.

Because our dataset has over 110 features, it would be more practical to narrow down the features to a few that we might be interested in. Through trial-and-error, we narrowed down features that would give us a fairly high Silhouette score (which ranges between -1 and 1).

We also narrowed the soccer position down to just Goalkeepers, because we want to narrow our scope and concentrate our effort on analyzing small set of data. Based on our model, top goalkeepers in the dataset are put in Cluster 2, and lowest ranking goalkeepers are put in Cluster 1.

We wanted our webapp to be able to take in real-time input from soccer scouts, where they would choose scores for different skill sets using sliders. However, we were faced with technical challenges and time constraint to execute such an app, so we had to scale down our original plan. Instead, after much discussion, we created an app that outputs players in different clusters using FIFA22 dataset. Unfortunately, we weren't able to implement real-time input functionality.

Clearly, we noticed that unsupervised learning requires our own interpretation of the data, as opposed to supervised learning, which we know the labels beforehand. We had to spend some time understanding why players are put in different clusters and which attributes were outweigh others. For example, **Overall score** attribute seem to affect the Silhouette score of our K-means model more than other attributes, such as age, reflexes, speed.

Finally, we were constrained by the shorter time to work on this assignment, as well as busy schedules of our group members.

7. Bibliography

1. D'Urso, P., De Giovanni, L. & Vitale, V. A robust method for clustering football players with mixed attributes. Ann Oper Res (2022). <https://doi.org/10.1007/s10479-022-04558-x>
2. Soccermetrics by David Sumpter
<https://soccermetrics.readthedocs.io/en/latest/index.html>
3. The Hundred Page Machine Learning Book. By Andriy Burkov

4. Parallel Coordinate Plots by Plotly <https://plotly.com/python/parallel-coordinates-plot/>